<u>Lab 14 - Open Ended Lab - Report</u>

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Course: Programming Fundamentals – Lab

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Task – 1

```
#include<iostream>
#include<string>
using namespace std;

struct Car
{
   string brand;
   int year;
   float price;
};

int main()
{
   Car c1;

   c1.brand = "Toyota";
   c1.year  = 2022;
   c1.price = 4500000.50;

   cout << "Car Brand: " << c1.brand << endl;
   cout << "Manufacturing Year: " << c1.year <<
endl;
   cout << "Car Price: " << c1.price << endl;
```

```
    return 0;
}
```

Output

```
PS E:\Lab Tasks\Lab Task 14 - Open Ended Lab> ./14.1.exe
Car Brand: Toyota
Manufacturing Year: 2022
Car Price: 4.5e+06
```

Task – 2

```
#include<iostream>
#include<string>
using namespace std;

struct Student
{
    string name;
    int rollNo;
    float marks;
};

int main()
{
    Student s [ 5 ];
```

```
 for ( int i = 0; i < 5; i++ )
{
   cout << "Enter details of student " << i + 1 <<
endl;

   cout << "Name: ";
   cin >> s [ i ].name;

   cout << "Roll Number: ";
   cin >> s [ i ].rollNo;

   cout << "Marks: ";
   cin >> s [ i ].marks;

   cout << endl;
}

cout << "Student Details" << endl;
cout << "----------------" << endl;

for ( int i = 0; i < 5; i++ )
{
   cout << "Student " << i + 1 << endl;
   cout << "Name: " << s [ i ].name << endl;
```

```
        cout << "Roll Number: " << s [ i ].rollNo << endl;
        cout << "Marks: " << s [ i ].marks << endl;
        cout << endl;
    }

    return 0;
}
```

Output

```
PS E:\Lab Tasks\Lab Task 14 - Open Ended Lab> ./14.2.exe
Enter details of student 1
Name: s1
Roll Number: 1
Marks: 37

Enter details of student 2
Name: s2
Roll Number: 2
Marks: 57

Enter details of student 3
Name: s3
Roll Number: 3
Marks: 45

Enter details of student 4
Name: s4
Roll Number: 4
Marks: 89

Enter details of student 5
Name: s5
Roll Number: 5
Marks: 97
```

```
Student Details
----------------
Student 1
Name: s1
Roll Number: 1
Marks: 37

Student 2
Name: s2
Roll Number: 2
Marks: 57

Student 3
Name: s3
Roll Number: 3
Marks: 45

Student 4
Name: s4
Roll Number: 4
Marks: 89

Student 5
Name: s5
Roll Number: 5
Marks: 97
```

Task – 3

```
#include<iostream>
using namespace std;

struct Rectangle
{
    float length;
    float breadth;
```

```
};

float calculate_area ( Rectangle r )
{
    float a = r.length * r.breadth;
    return a;
}

float calculate_perimeter ( Rectangle r )
{
    float p = 2 * ( r.length + r.breadth );
    return p;
}

int main()
{
    Rectangle rect;

    cout << "Enter Length: " << endl;
    cin >> rect.length;

    cout << "Enter Breadth: " << endl;
    cin >> rect.breadth;

    float area = calculate_area ( rect );
```

```
    float perimeter = calculate_perimeter ( rect );

    cout << "Area of Rectangle: " << area << endl;
    cout << "Perimeter of Rectangle: " << perimeter
<< endl;

    return 0;
}
```

Output

```
PS E:\Lab Tasks\Lab Task 14 - Open Ended Lab> ./14.3.exe
Enter Length:
3.7
Enter Breadth:
8.9
Area of Rectangle: 32.93
Perimeter of Rectangle: 25.2
```

Task – 4

```
#include <iostream>
#include <string>

using namespace std;

struct Book
{
```

```
    string title;
    string author;
    float price;
};

int main()
{
    Book b1, b2;

    cout << "Enter details of first book" << endl;

    cout << "Title: ";
    getline ( cin, b1.title );

    cout << "Author: ";
    getline ( cin, b1.author );

    cout << "Price: ";
    cin >> b1.price;

    cin.ignore();

    cout << endl;

    cout << "Enter details of second book" << endl;
```

```
cout << "Title: ";
getline ( cin, b2.title );

cout << "Author: ";
getline ( cin, b2.author );

cout << "Price: ";
cin >> b2.price;

cout << endl;

if ( b1.price > b2.price )
{
    cout << "Book with higher price:" << endl;
    cout << "Title: " << b1.title << endl;
    cout << "Author: " << b1.author << endl;
    cout << "Price: " << b1.price << endl;
}
else if ( b2.price > b1.price )
{
    cout << "Book with higher price:" << endl;
    cout << "Title: " << b2.title << endl;
    cout << "Author: " << b2.author << endl;
    cout << "Price: " << b2.price << endl;
```

```
    }
    else
    {
        cout << "Both books have the same price." << endl;
    }

    return 0;
}
```

Output

```
PS E:\Lab Tasks\Lab Task 14 - Open Ended Lab> ./14.4.exe
Enter details of first book
Title: The Kite Runner
Author: Khaled Hosseini
Price: 1500

Enter details of second book
Title: Peer - e - Kamil
Author: Umera Ahmed
Price: 5000

Book with higher price:
Title: Peer - e - Kamil
Author: Umera Ahmed
Price: 5000
```

THE END